

AI-compute supply chain: chart pack

Current state of the analysis. Data: UN Comtrade + TDM monthly panel (60 HS6 codes, 2017-01..2026-04, audited vs Atlas) and Atlas HS2012 annual.
Sections: 1 the chain as flows (2024) -- 2 the time-varying factor model -- 3 concentration & fragmentation.
Details: docs/data.md, docs/supply-chain-narrative.md, results/.

2026-07-26

1. The chain as flows

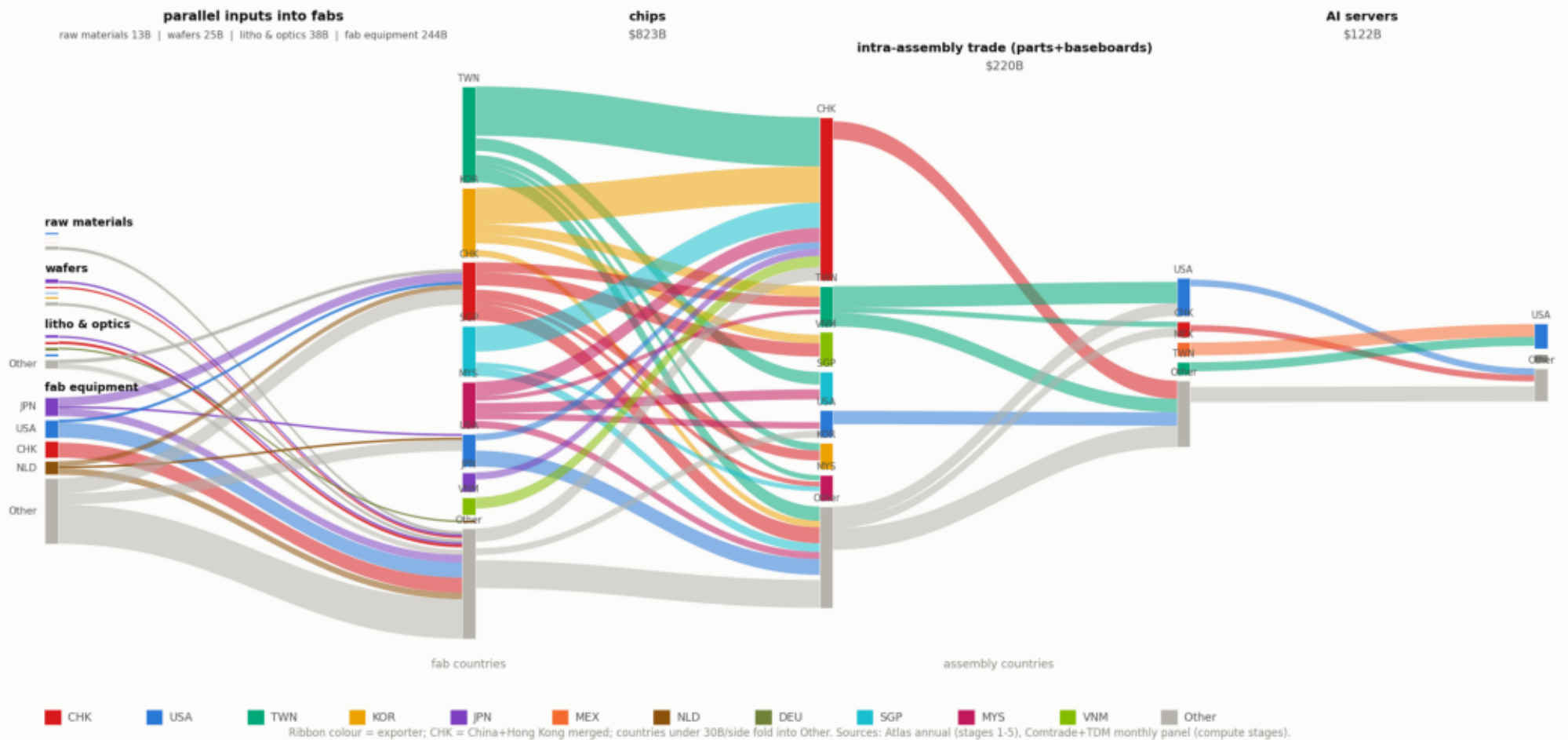
2024 bilateral flows, China+Hong Kong merged (CHK), intra-bloc excluded. Stages 1-5 from Atlas HS2012 annual data; stages 6-8 from the monthly panel (docs/data.md §3). Generated by scripts/export_supply_chain_sankey.py.

Whole-chain overview, dollar scale (coarse)

1. The chain as flows

All eight stages left to right; band width = 2024 dollars. Inputs, litho optics and equipment run in parallel and converge on the fab countries; parts and baseboards form the intra-assembly segment before servers.

The AI-compute supply chain, 2024 — parallel inputs converge on the fabs; chips flow on to servers (one dollar scale — band widths nominally comparable everywhere)

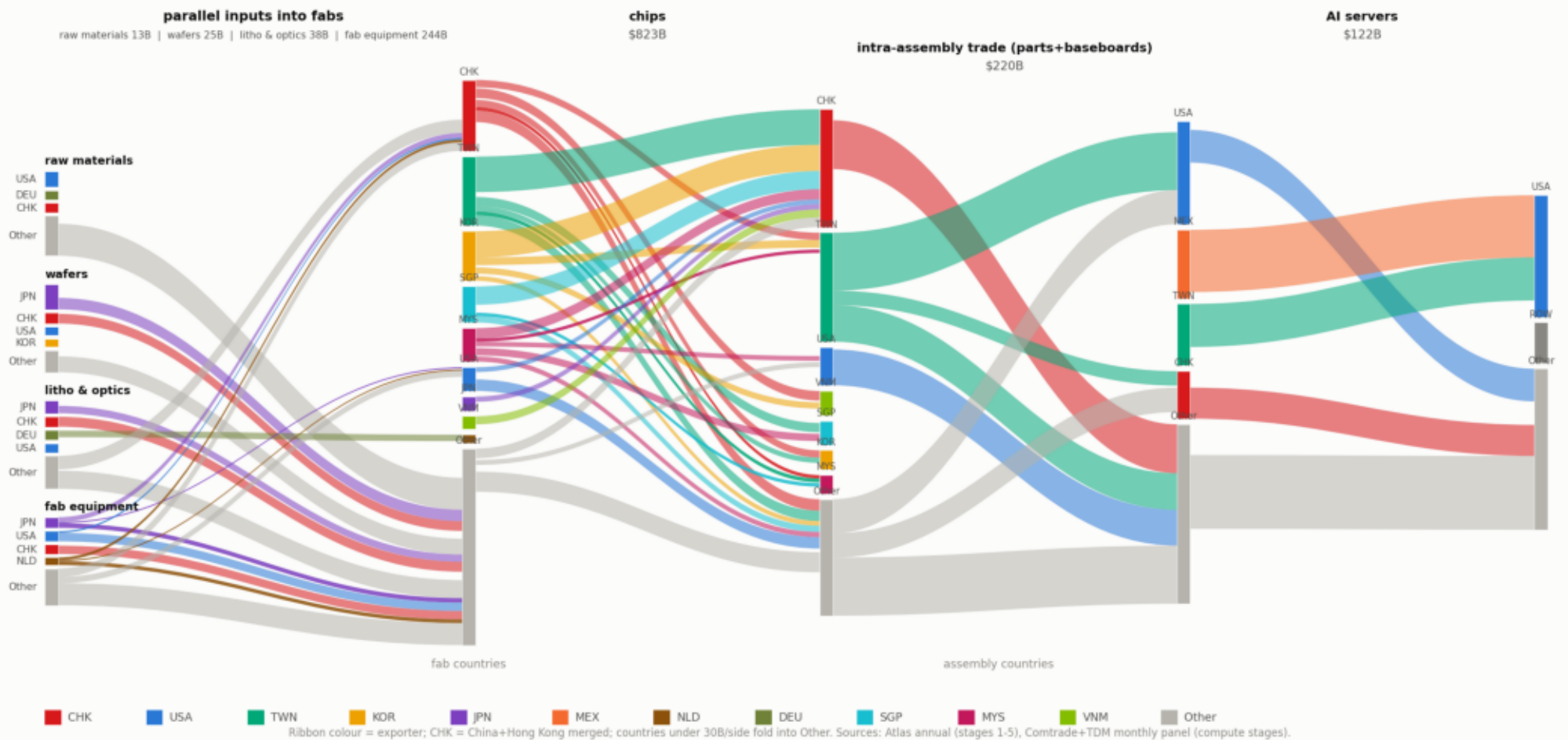


Whole-chain overview, normalized (coarse)

1. The chain as flows

Same flows with each stage rescaled to equal width, so small stages' composition is readable (dollar scale makes chips dwarf everything).

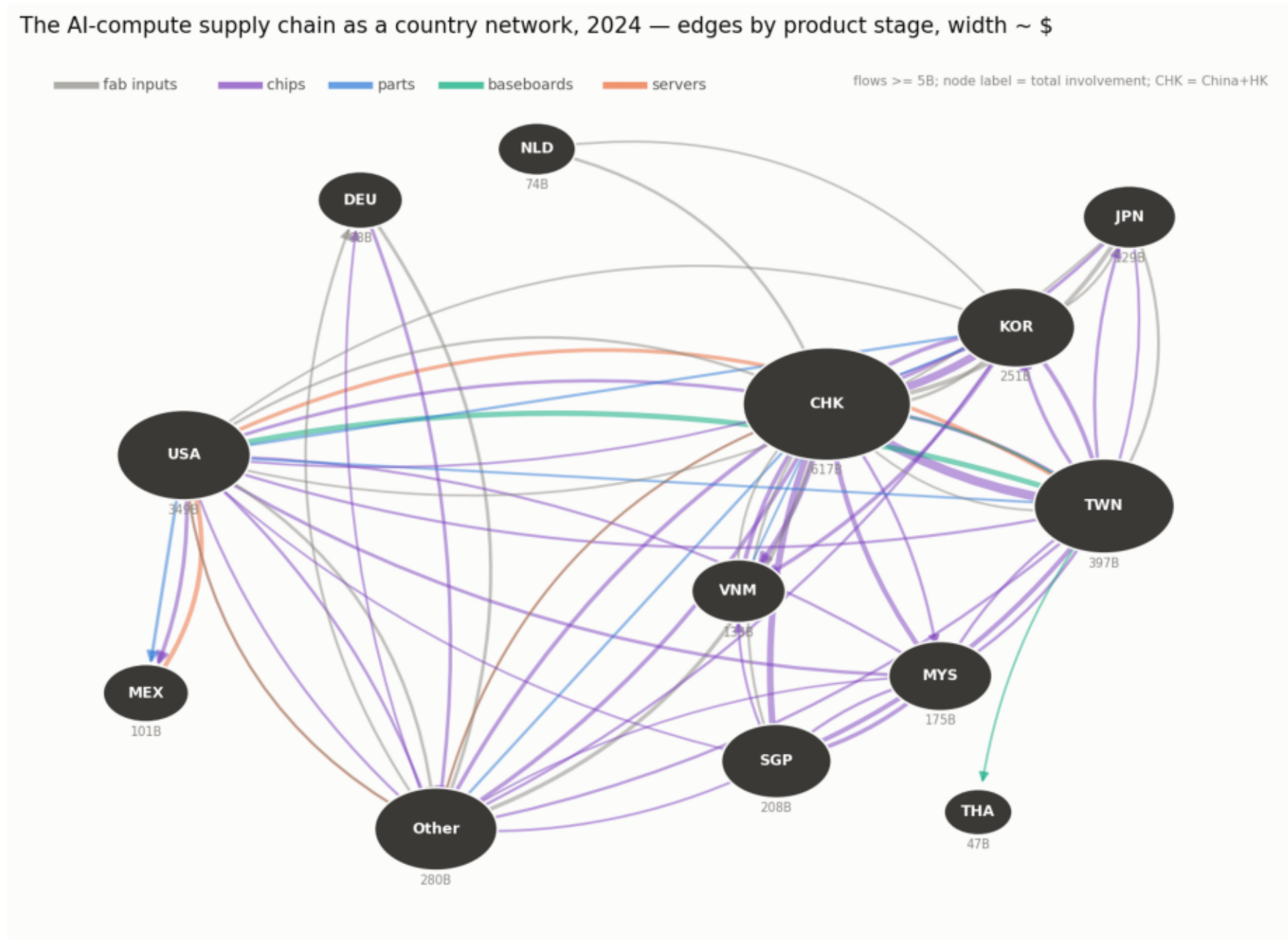
The AI-compute supply chain, 2024 — parallel inputs converge on the fabs; chips flow on to servers (stages equalized — shares only)



Country network, all stages

1. The chain as flows

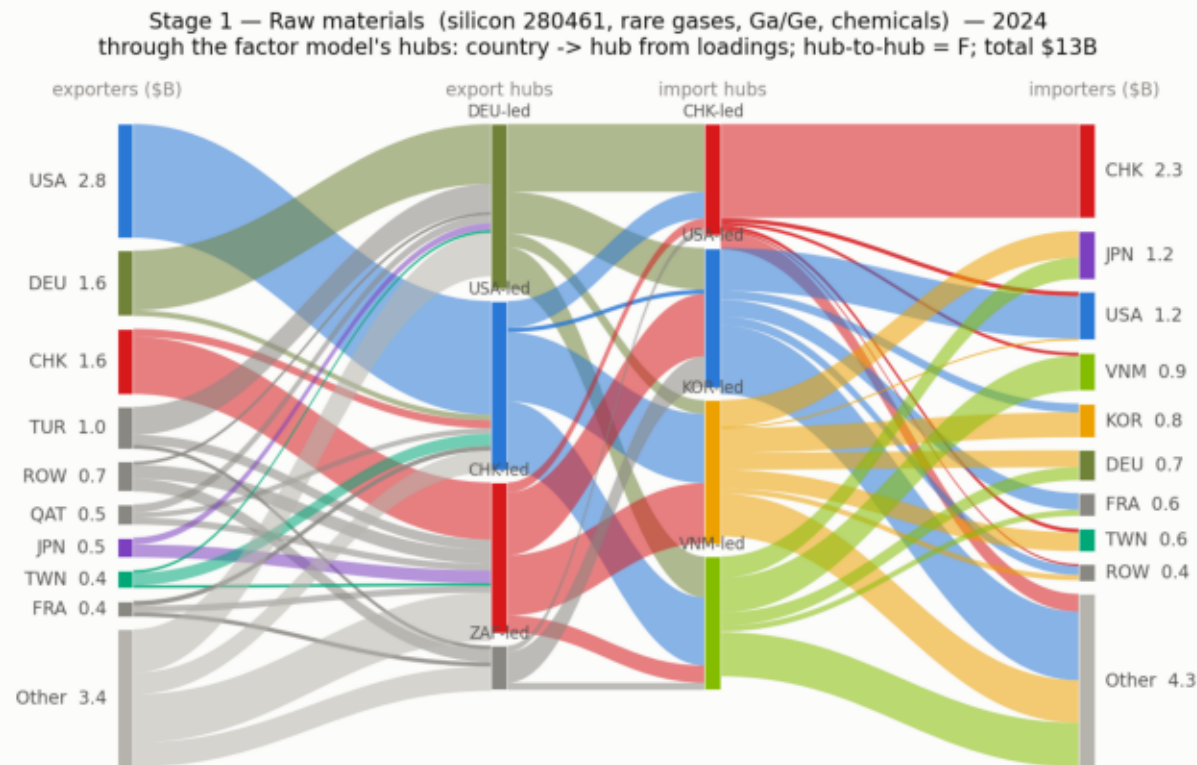
Nodes = countries, edges = 2024 flows colored by stage. Shows integrated roles a stage sequence cannot: Mexico takes in chips and parts and ships out servers at a single node.



Stage 1 -- raw materials, hub decomposition

1. The chain as flows

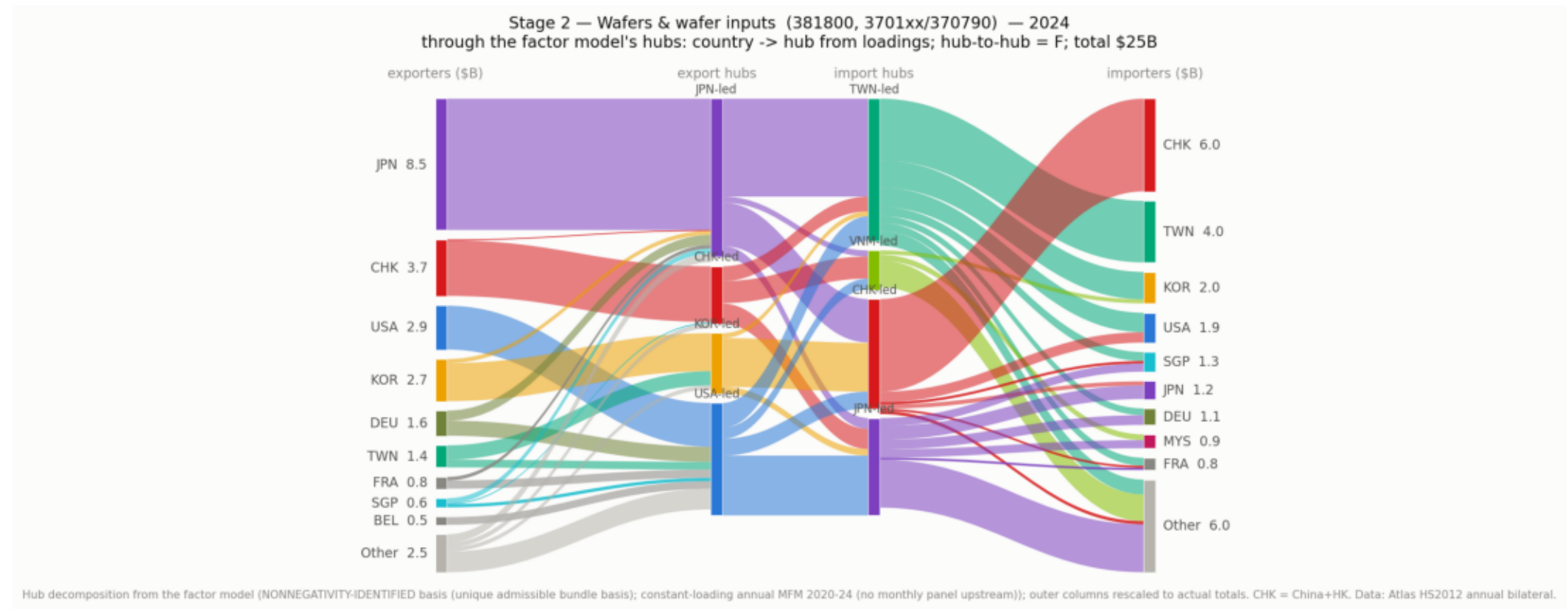
Four-column view: exporters -> export hubs -> import hubs -> importers, hubs estimated by MFM in the nonnegativity-identified basis. Applies to every stage page that follows; fits R^2 0.95-0.99.



Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.

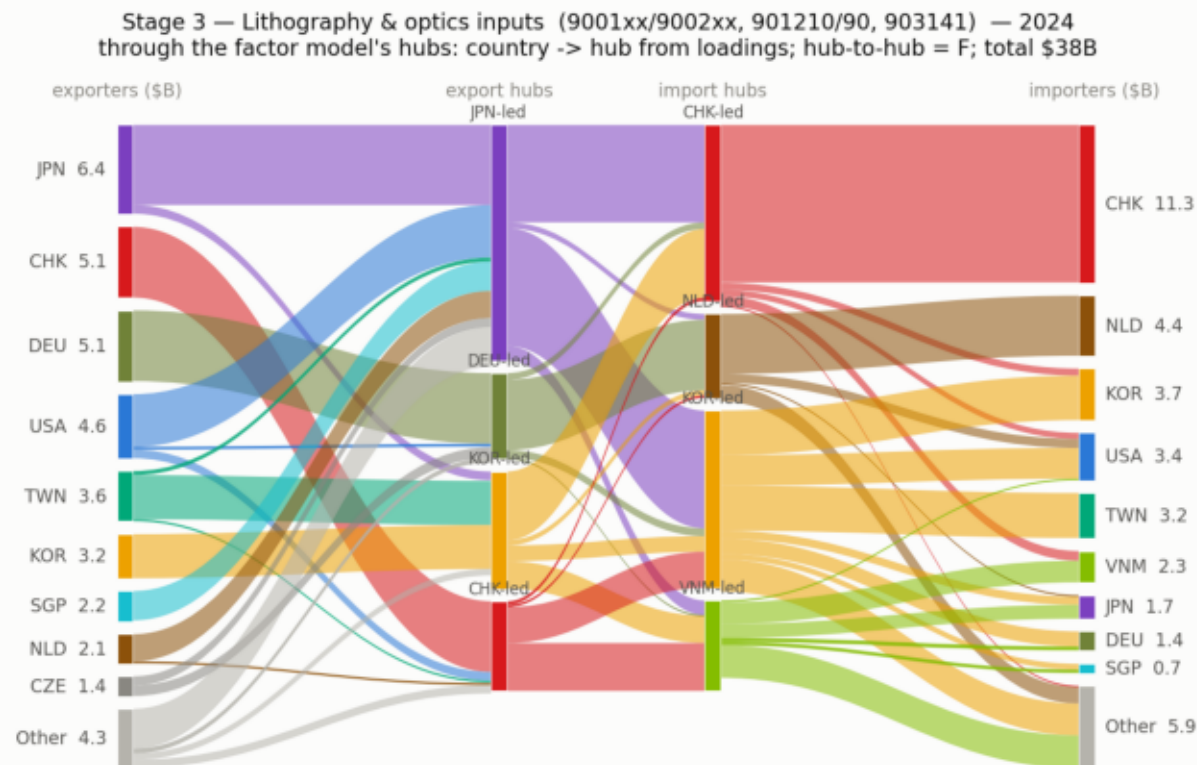
Stage 2 -- wafers & wafer inputs

1. The chain as flows



Stage 3 -- lithography & optics inputs

1. The chain as flows

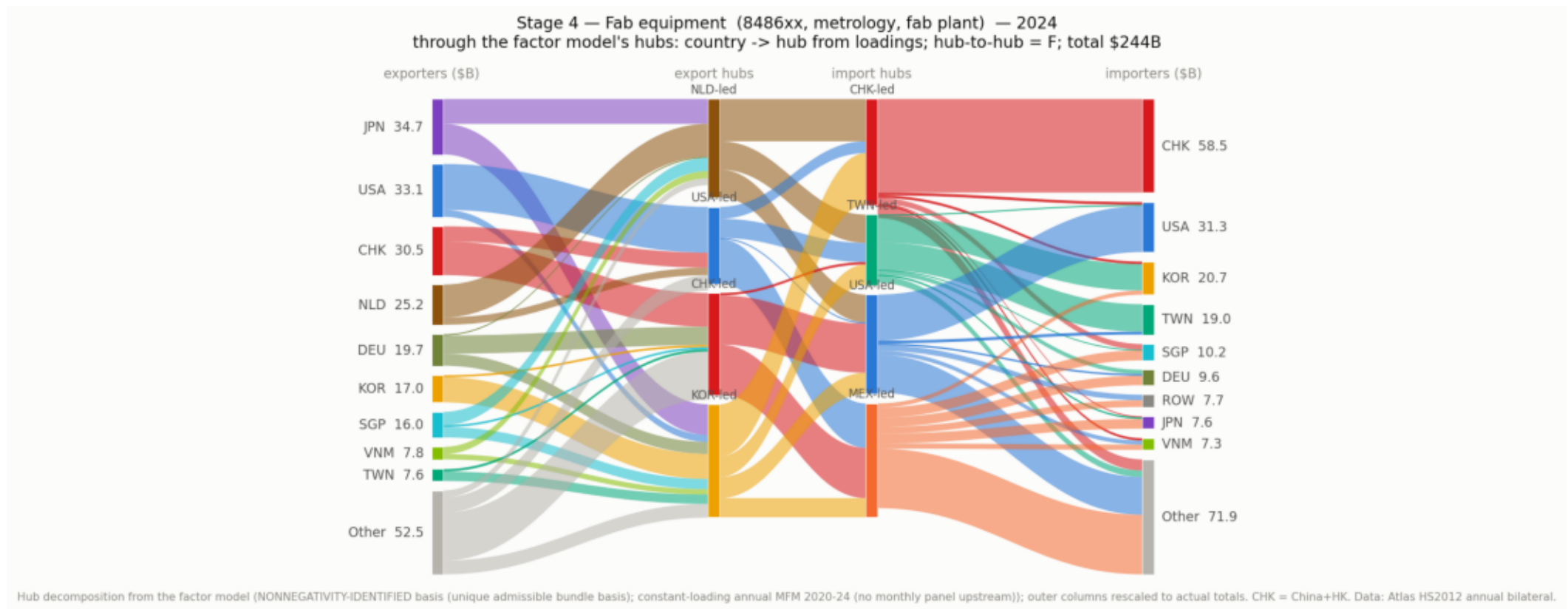


Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.

Stage 4 -- fab equipment

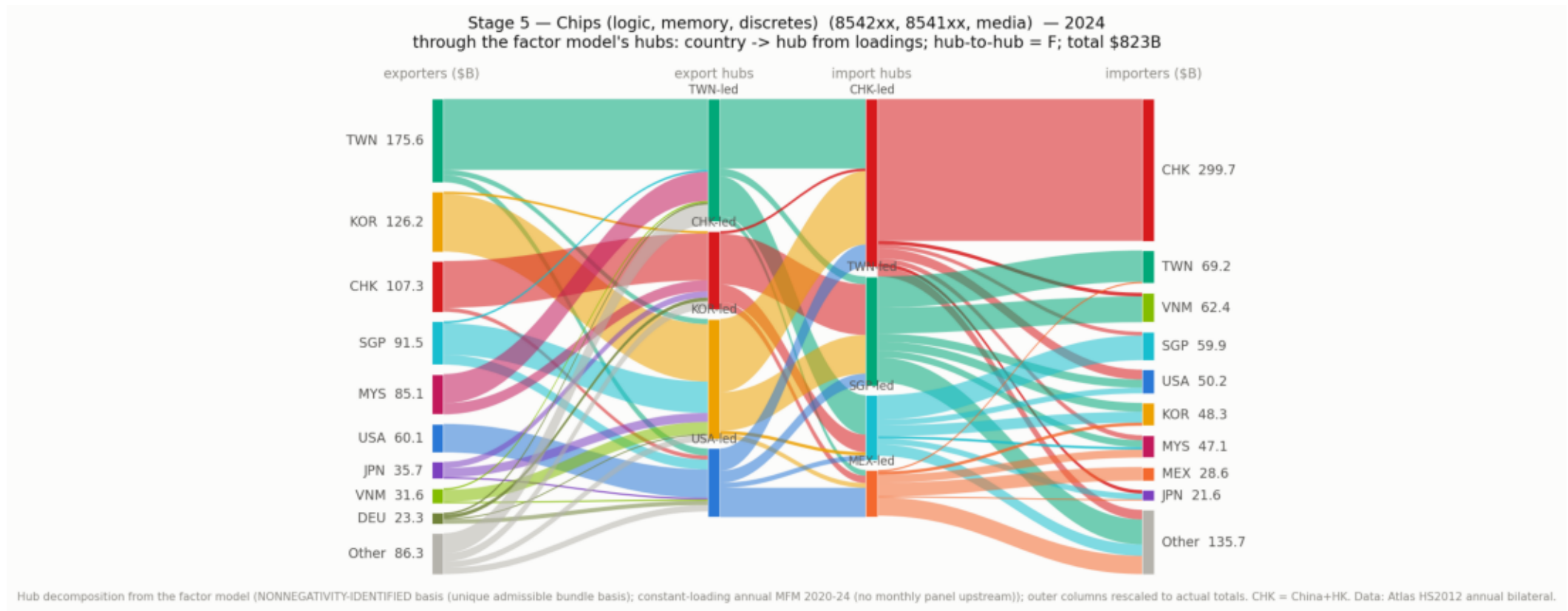
1. The chain as flows

Capacity goods, not consumables: in fab countries these imports lead chip exports by 6-12 months (docs/notes/edge-type-tests.md).



Stage 5 -- chips (logic, memory, discretes)

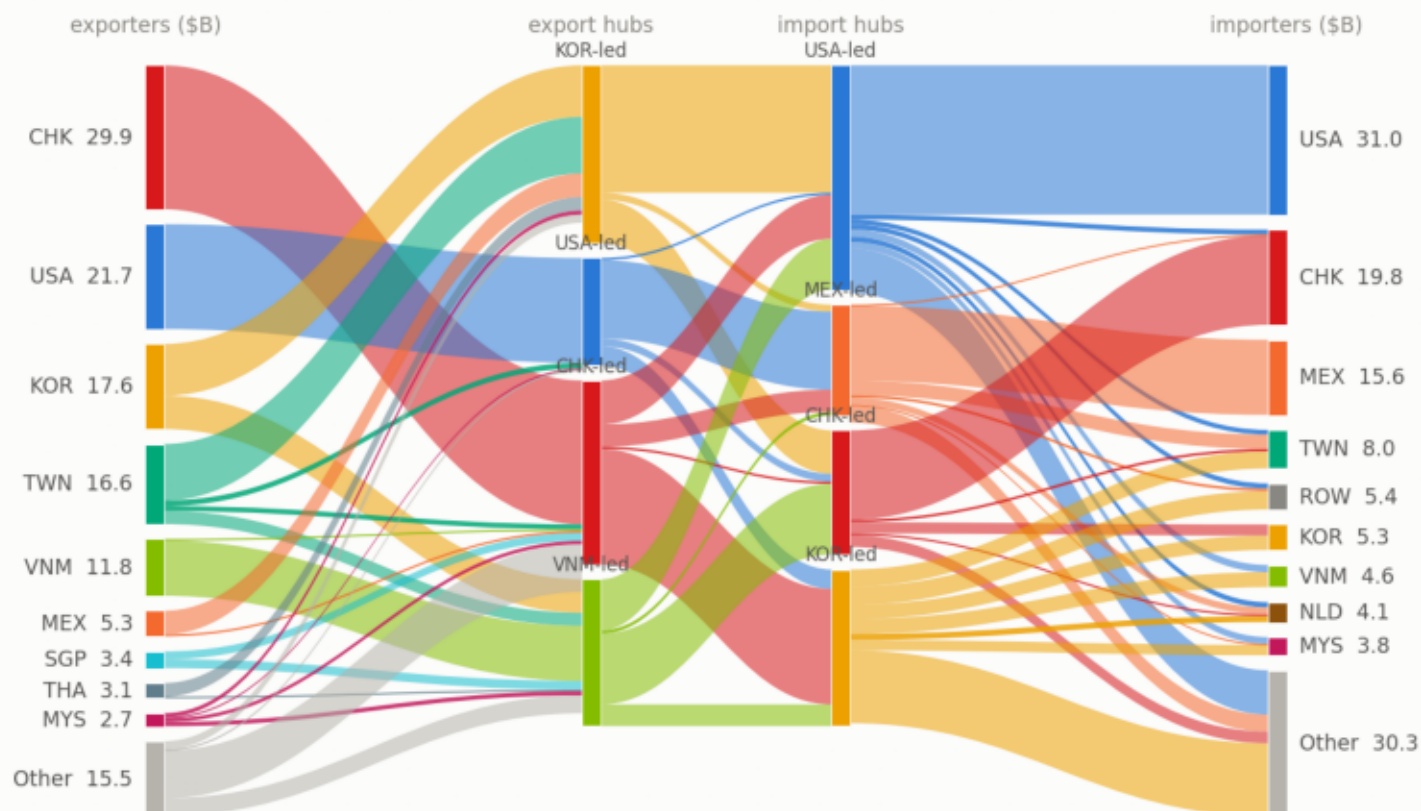
1. The chain as flows



Stage 6 -- parts & GPU modules (847330)

1. The chain as flows

Stage 6 — Parts & GPU modules (HS 847330) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$128B

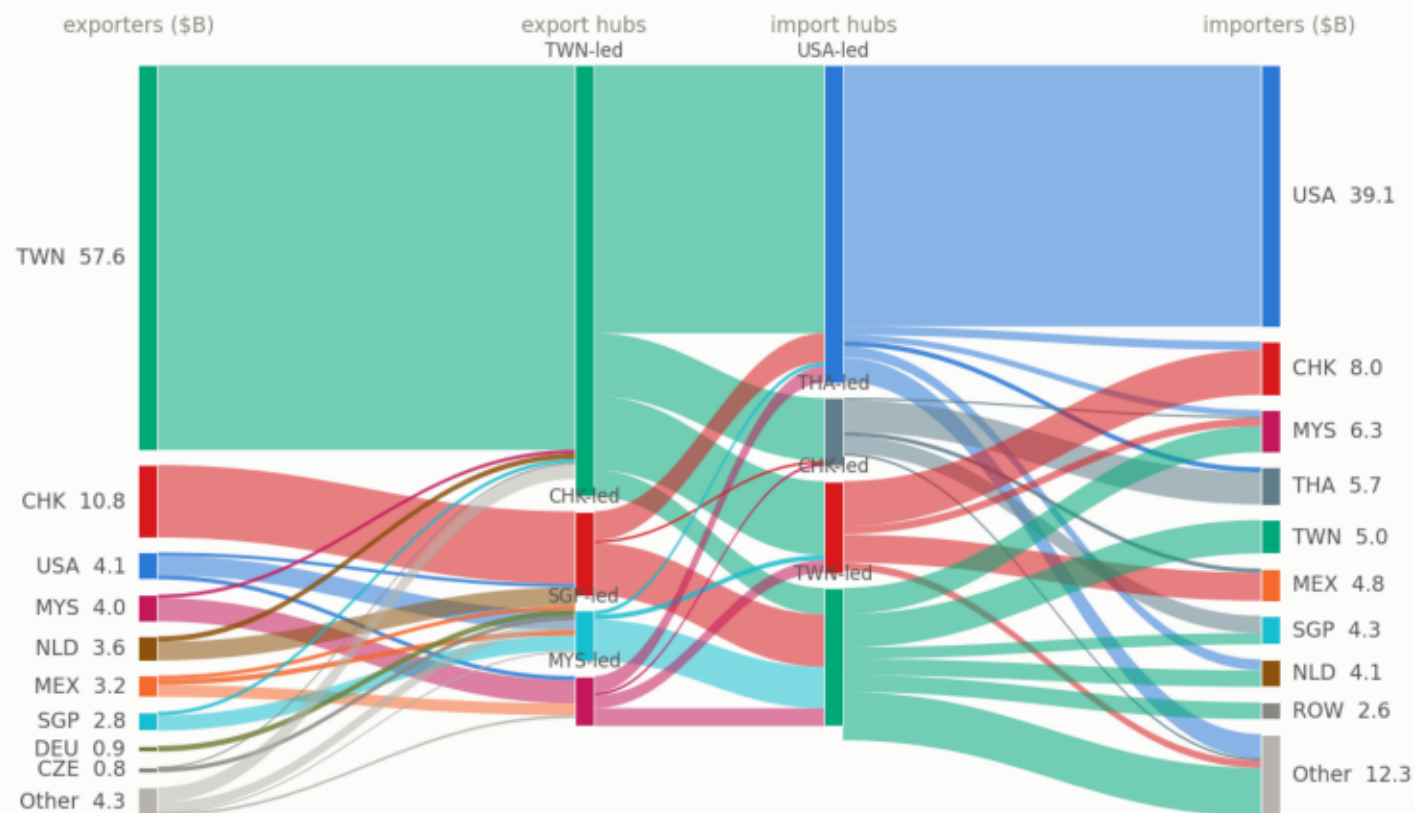


Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); pooled 2024 monthly MFM); outer columns rescaled to actual totals. CHK = China+HK. Data: Comtrade+TDM monthly panel.

Stage 7 -- baseboards (847180)

1. The chain as flows

Stage 7 — Baseboards / other units (HS 847180) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$92B

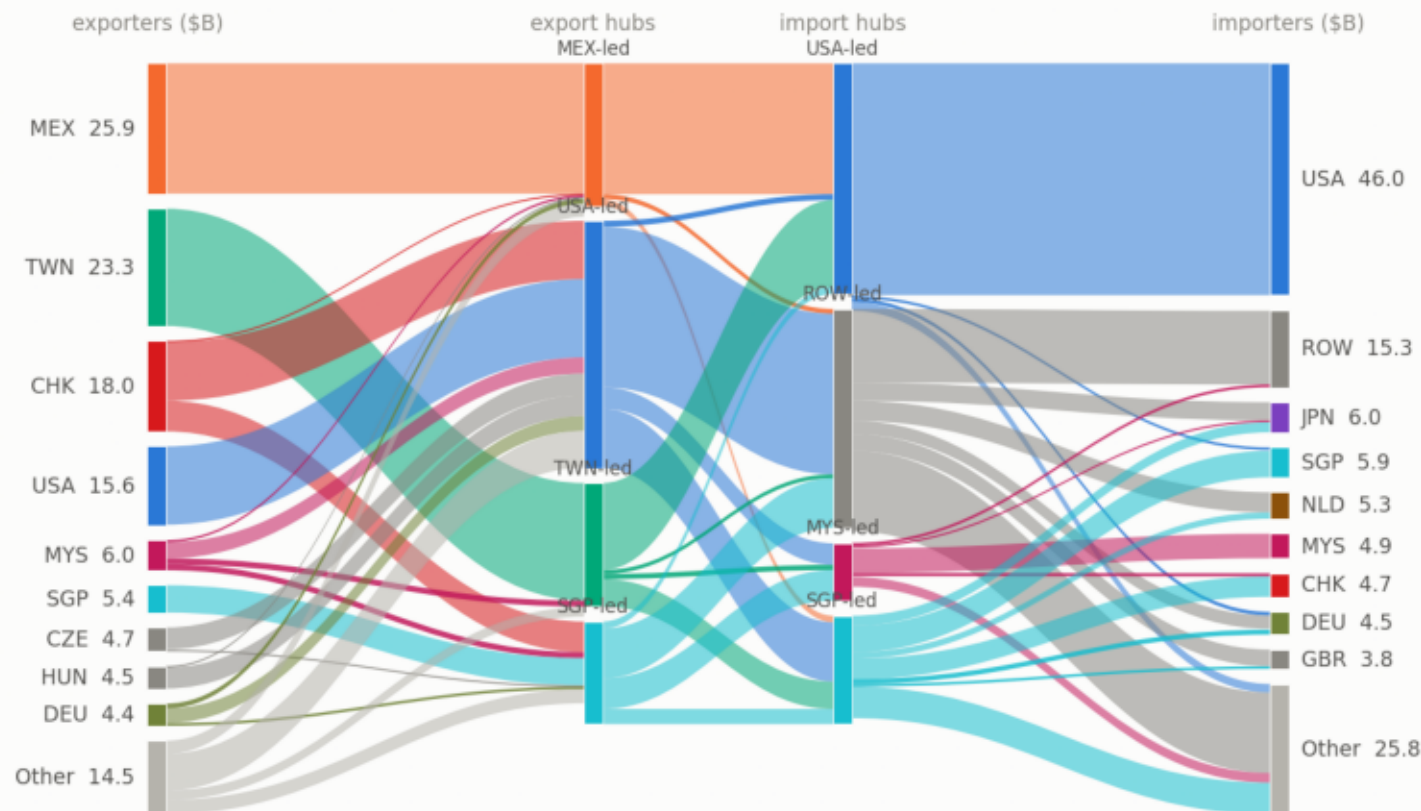


Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); pooled 2024 monthly MFM); outer columns rescaled to actual totals. CHK = China+HK. Data: Comtrade+TDM monthly panel.

Stage 8 -- AI servers (847150)

1. The chain as flows

Stage 8 — Finished AI servers (HS 847150) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$122B



Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); pooled 2024 monthly MFM); outer columns rescaled to actual totals. CHK = China+HK. Data: Comtrade+TDM monthly panel.

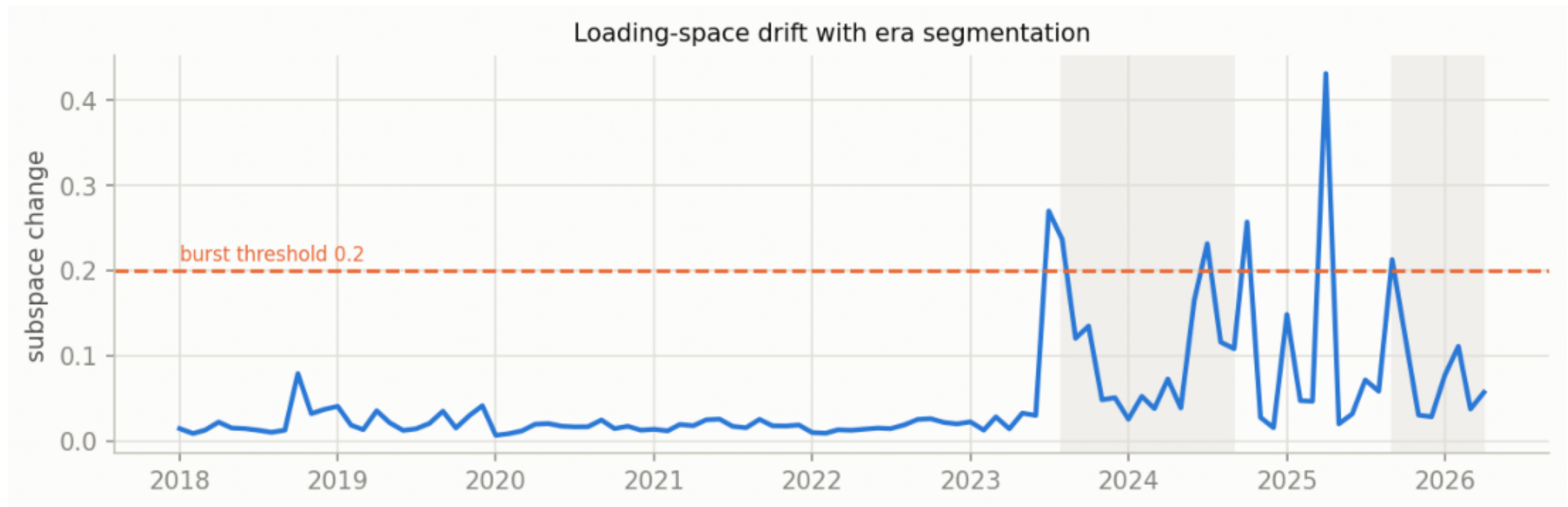
2. The time-varying factor model

Era-anchored TV-MFM on the audited monthly panel (12m trailing window, $k=4$, \$B levels, NNF basis), 2017-12..2026-04. Shaded bands = eras between structural breaks. Generated by `scripts/tvmfm_monthly_anchored.py`; summaries in `results/mfm/tvmfm/`.

AI compute, CHK basis -- loading drift and eras

2. The time-varying factor model

The preferred basis (entrepot churn netted; Taiwan separate). One unbroken 68-month era from 2017-12, then breaks at 2023-08, 2024-10, 2025-09.

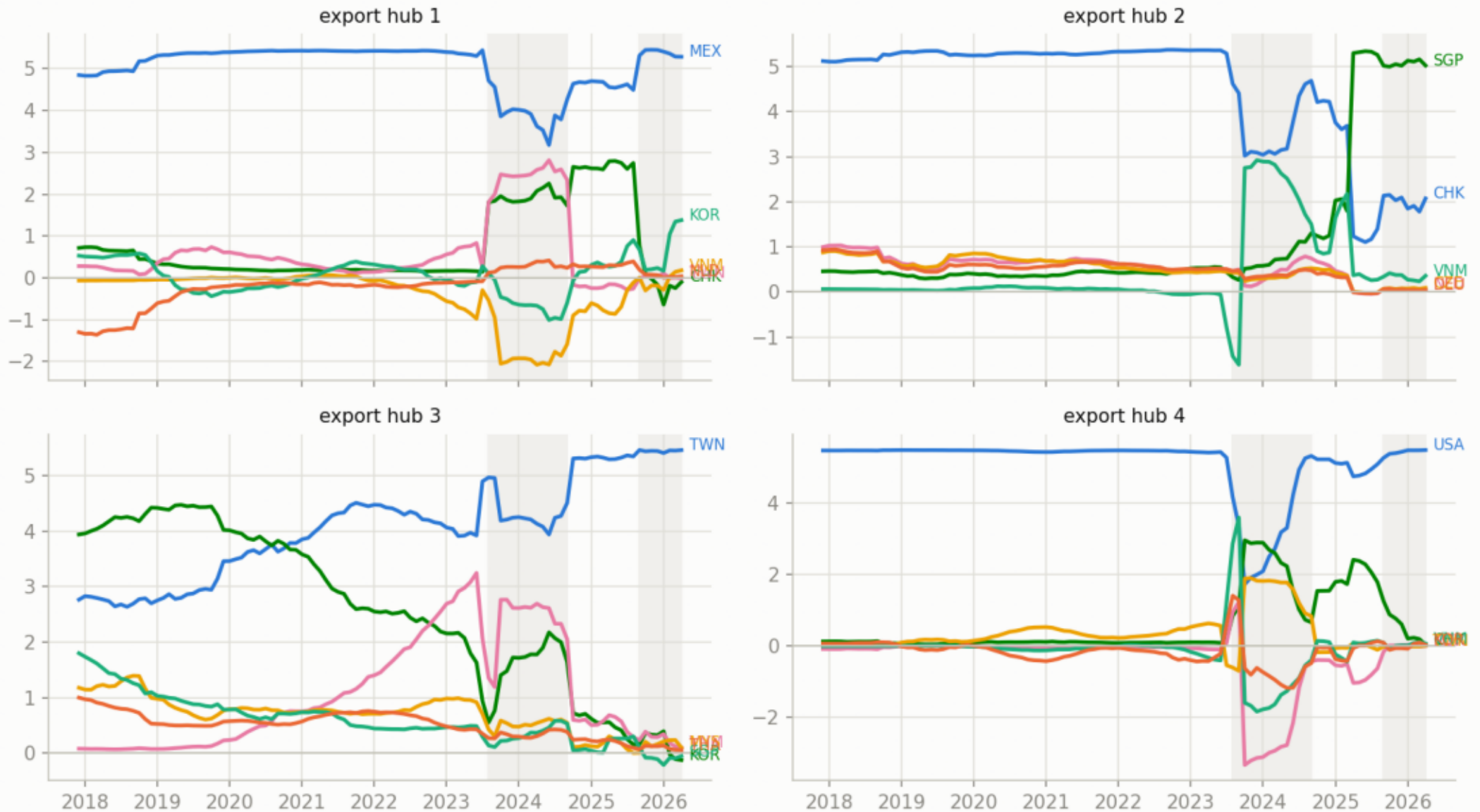


AI compute, CHK basis -- export-hub loadings

2. The time-varying factor model

Per-hub country loadings through time. At 2024-10 the China-bloc hub loses its separate identity (cross-era similarity 0.45) and a Singapore-led hub appears.

AI compute, China+HKG bloc (TWN separate) — export loadings, era-anchored (12m window; shaded bands = eras)

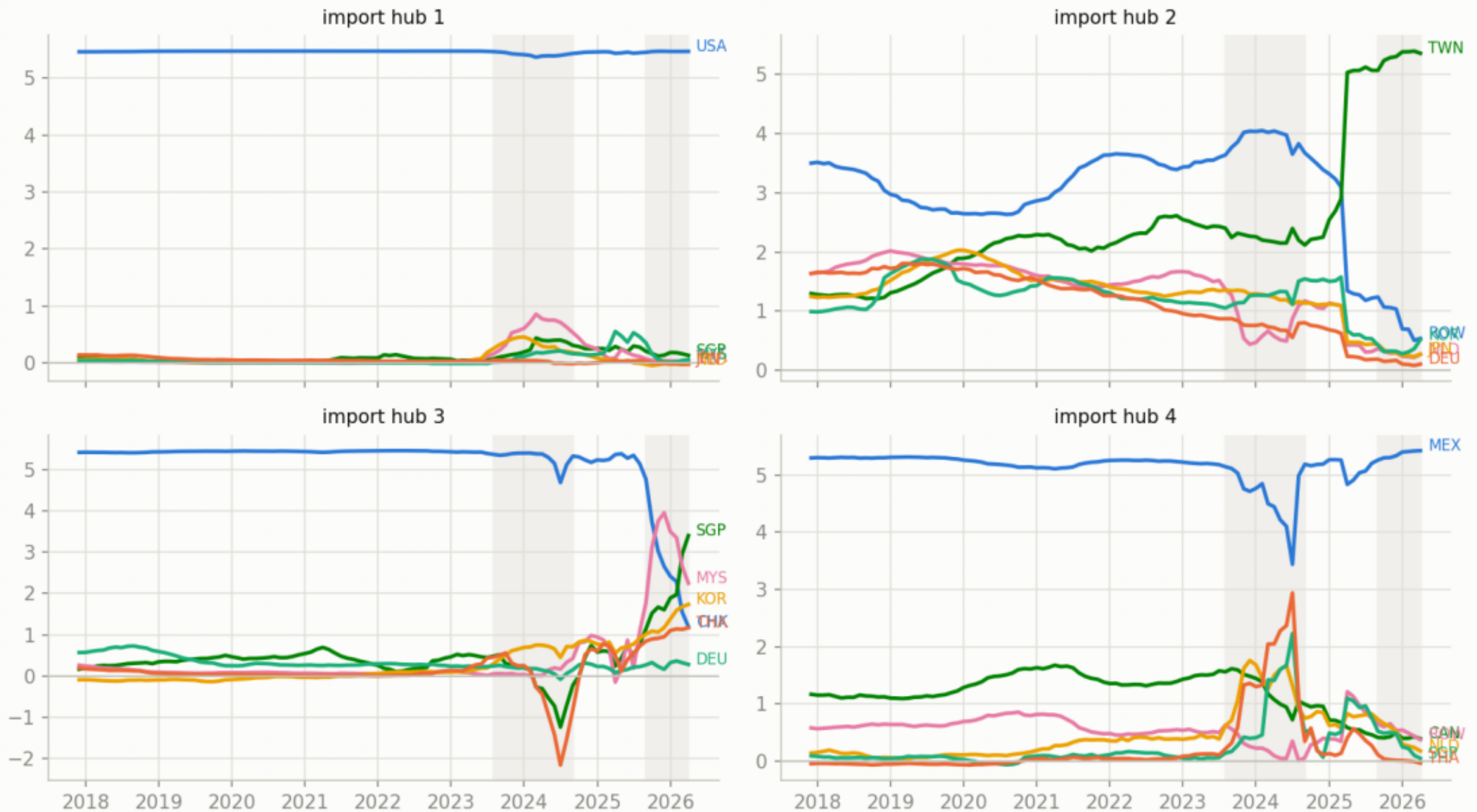


AI compute, CHK basis -- import-hub loadings

2. The time-varying factor model

Import hubs are the stable side of the network (USA, Mexico, Taiwan, rest-of-world led).

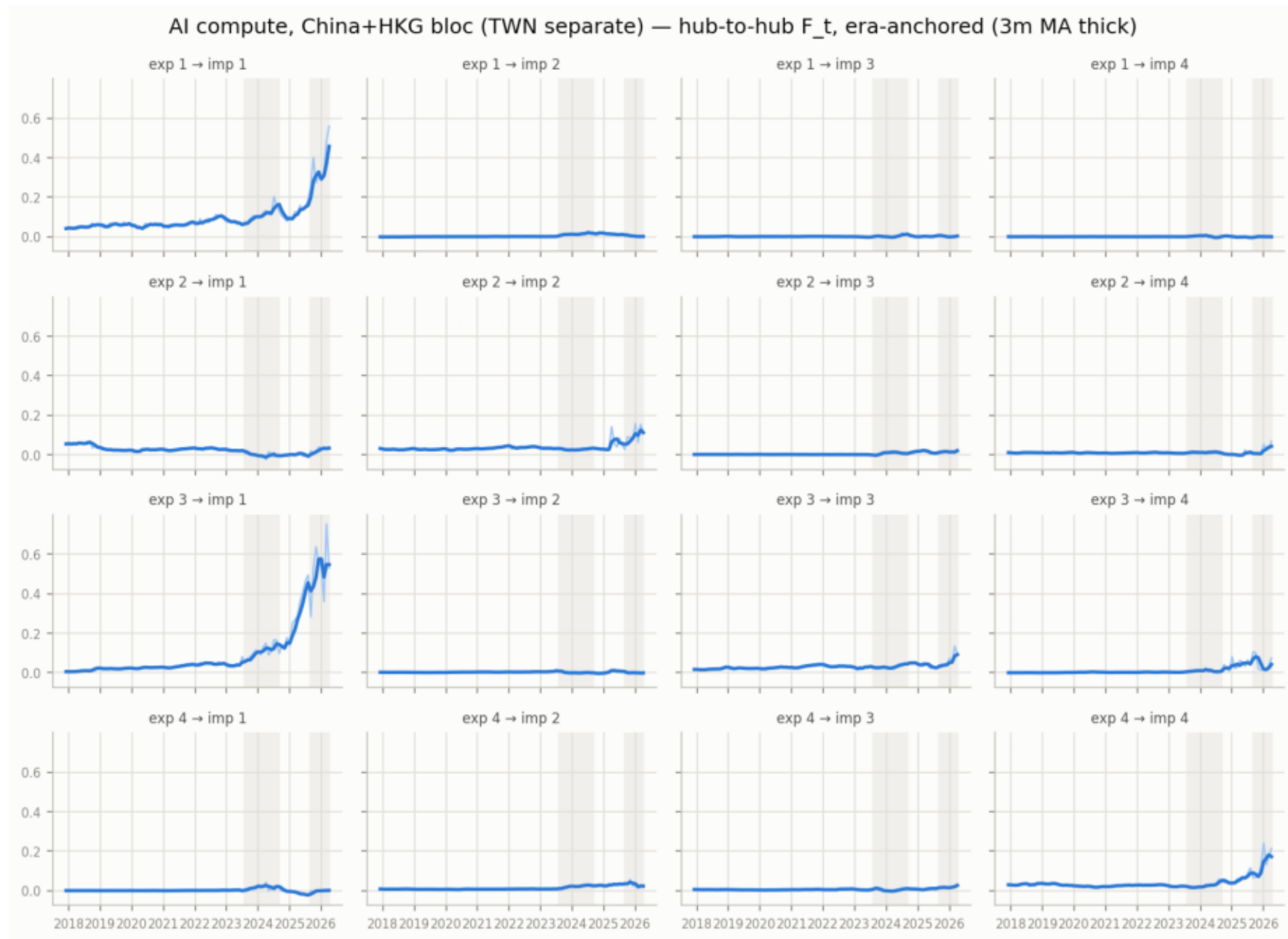
AI compute, China+HKG bloc (TWN separate) — import loadings, era-anchored (12m window; shaded bands = eras)



AI compute, CHK basis -- hub-to-hub intensities F_t

2. The time-varying factor model

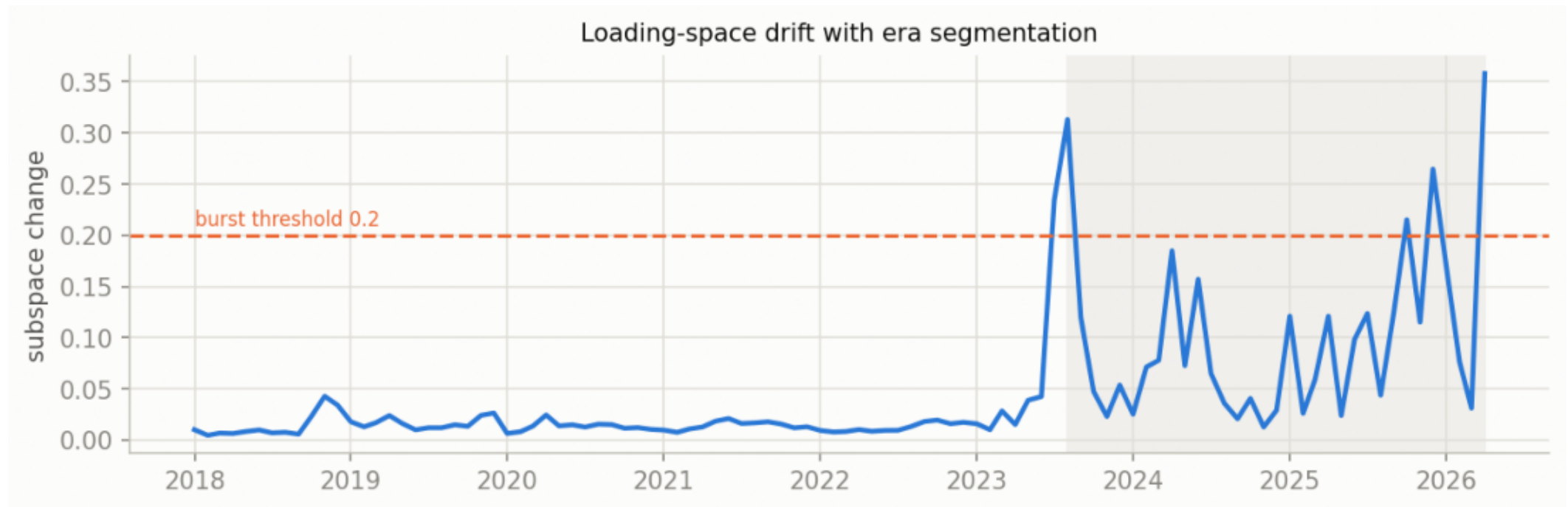
Each panel is one export-hub x import-hub channel in \$B (3m MA thick).



AI compute, country level -- drift

2. The time-varying factor model

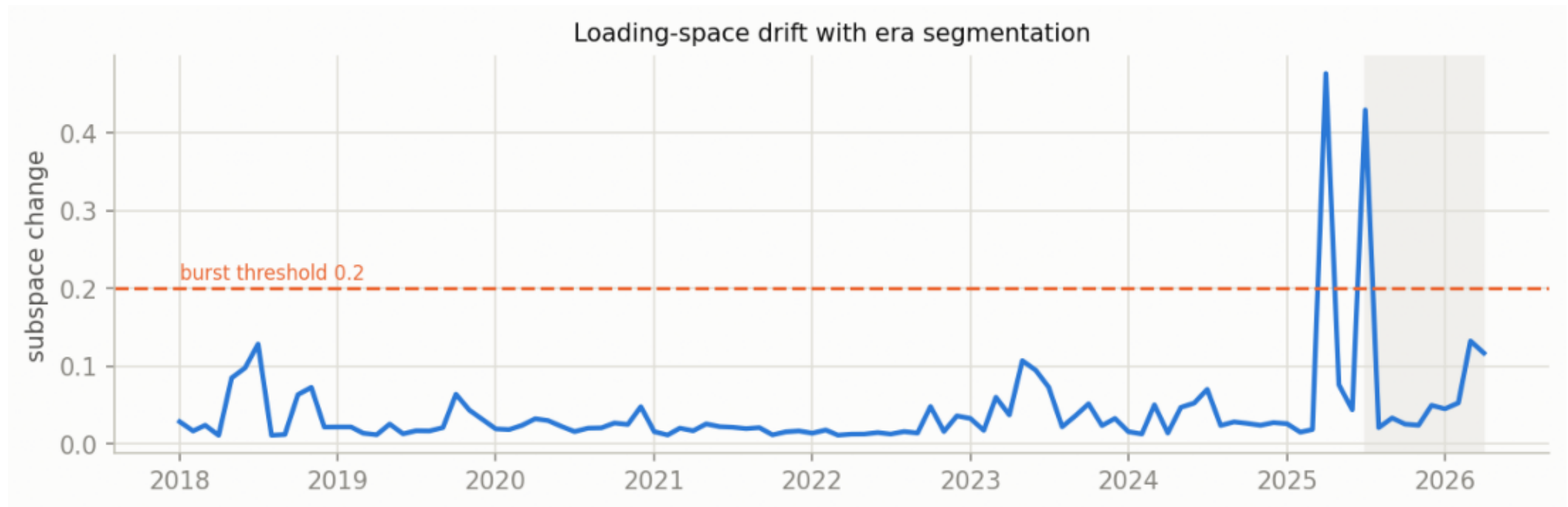
Without bloc merging the sample splits once: 2023-08. The demand event is the only dominant country-level break in nine years.



Parts (847330), country level -- drift

2. The time-varying factor model

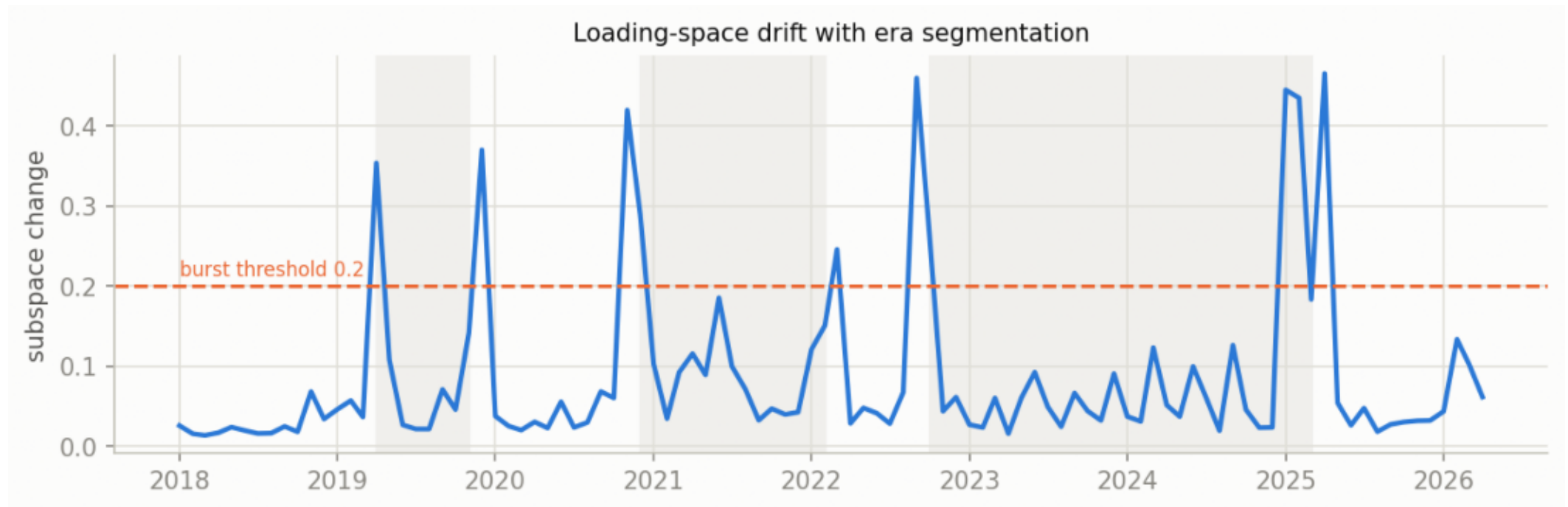
91 months without a structural break; the parts layer is the stable substrate of the network.



AI compute, double-bloc basis -- drift

2. The time-varying factor model

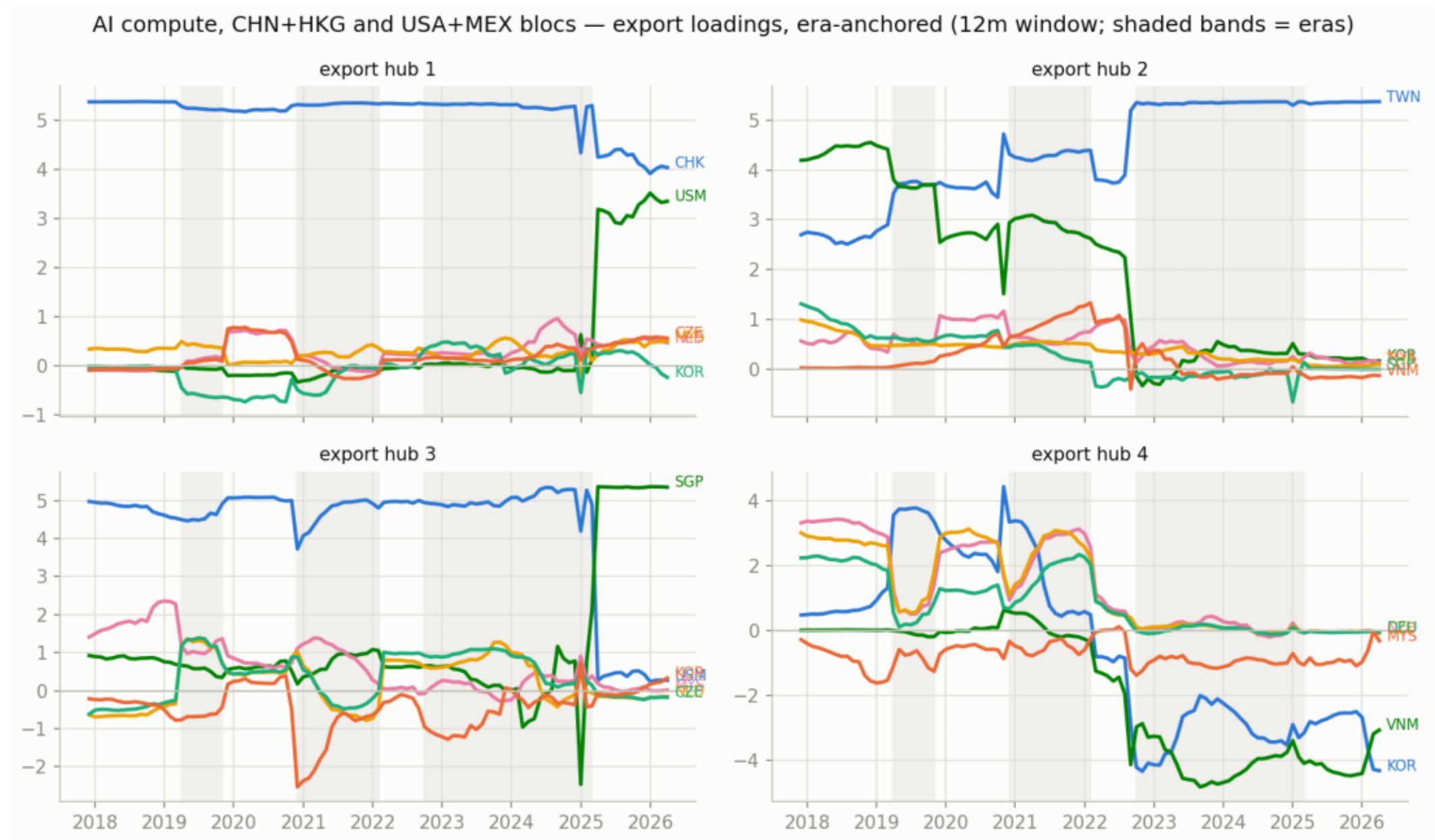
CHK and USA+MEX both merged, isolating cross-bloc structure. Breaks at 2019-04/2019-12 (first tariff war), 2022-10 (first export controls), 2025-04 (tariffs): policy breaks blocs.



AI compute, double-bloc basis -- export-hub loadings

2. The time-varying factor model

In the 2025-04 era the two blocs load together on one factor (CHK +3.97, USM +3.41) -- the entanglement signature unique to the tariff break.



3. Concentration & fragmentation

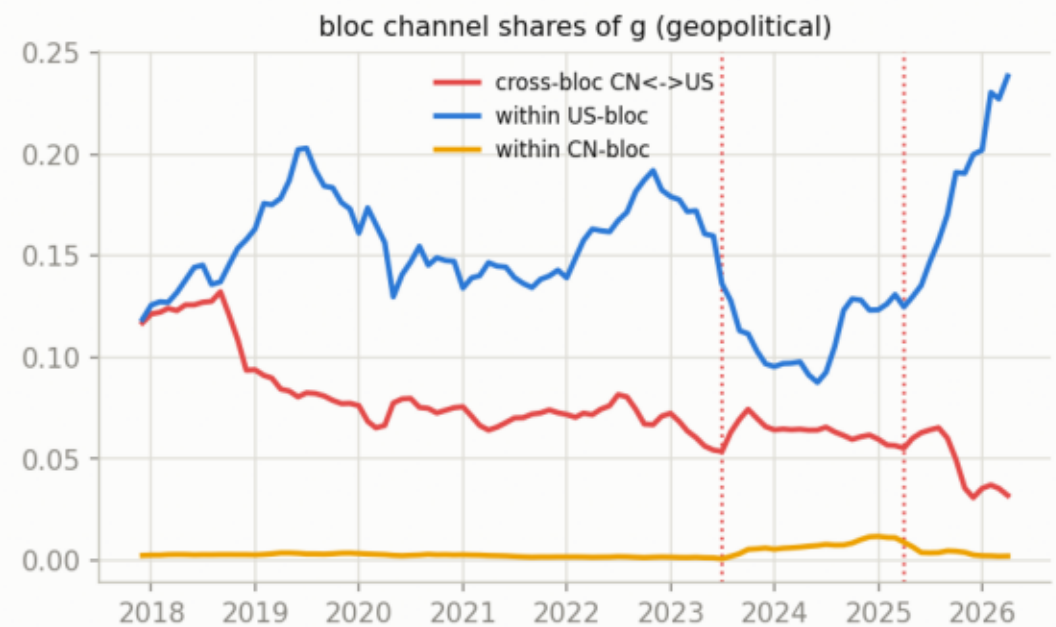
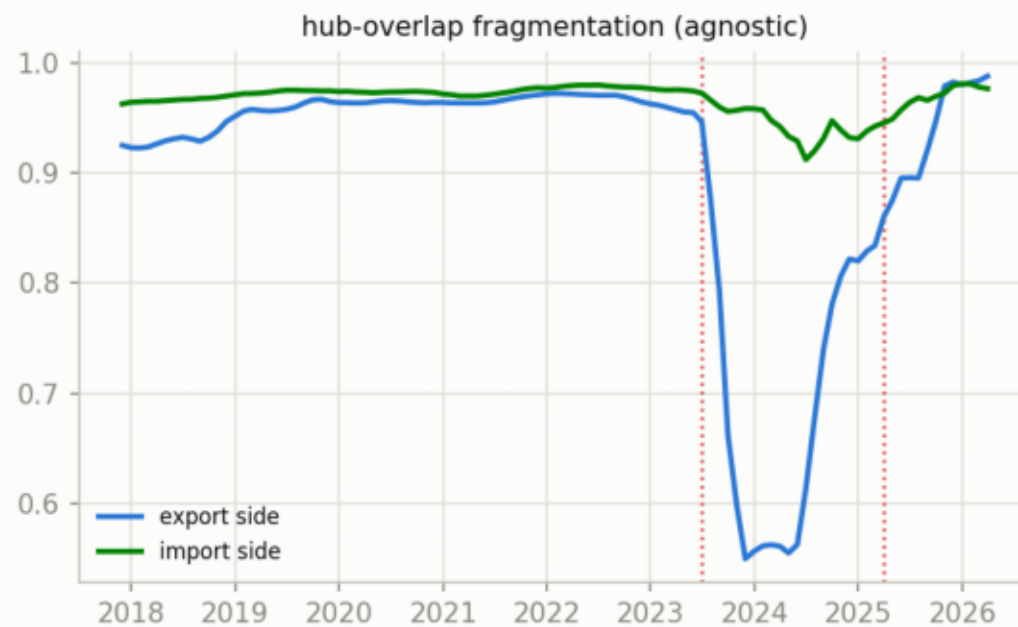
Measure system of docs/concentration-fragmentation-nnf-mfm-trade.md computed monthly from the TV-MFM (CHK basis). Dotted lines mark 2023-07 and 2025-04. Interpretation: results/network_stats/findings.md. Generated by scripts/network_stats.py.

Two notions of fragmentation -- AI compute

3. Concentration & fragmentation

Left: hub-overlap fragmentation (range-compressed under the anchored basis; mainly a diagnostic). Right: bloc channel shares -- the cross-bloc CN-US share falls ~70% over the sample in two legs (2019, then 2023 onward) with a plateau between.

AI compute (3 codes) — two notions of fragmentation (3m MA)

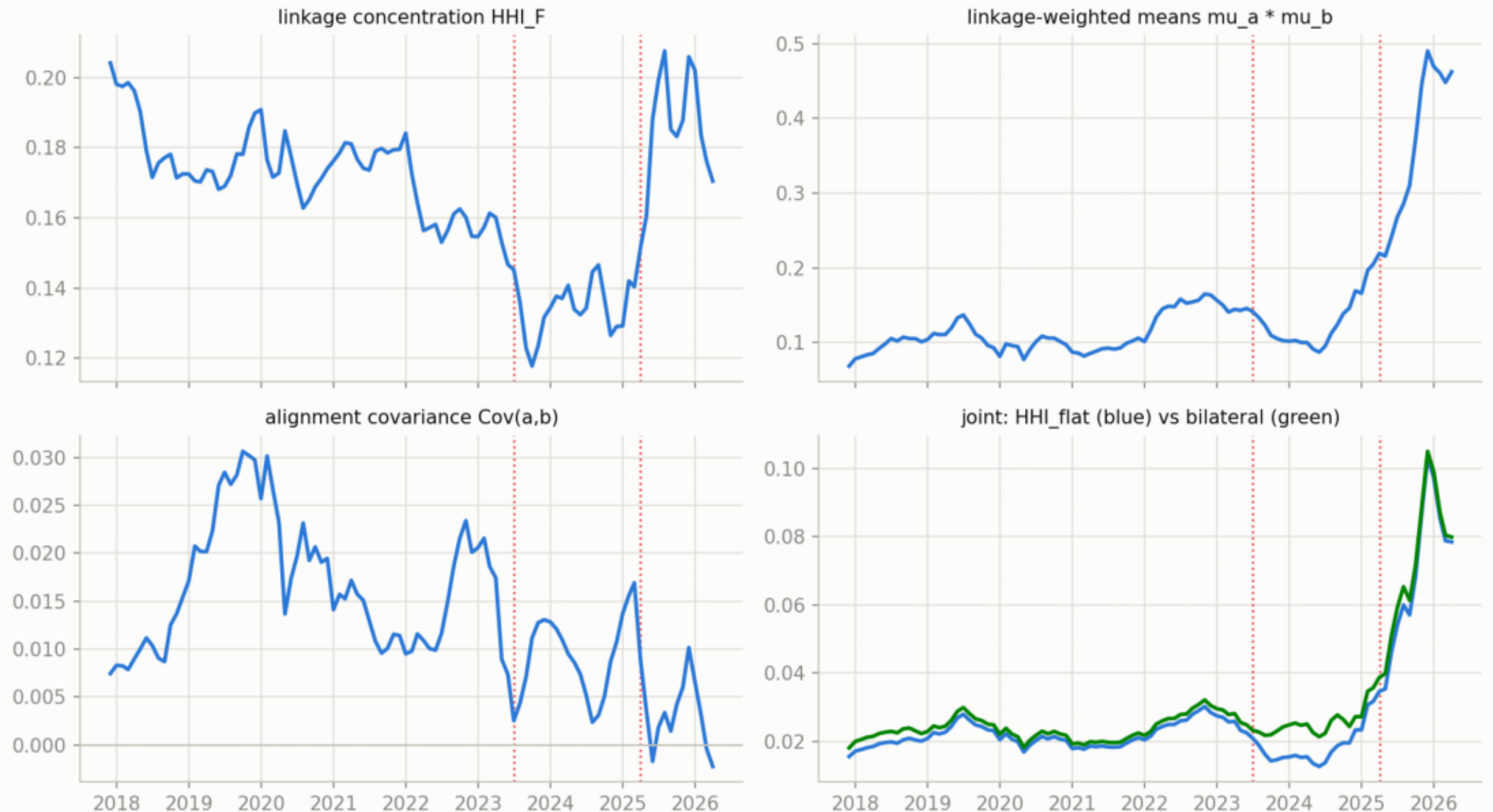


Joint concentration decomposition -- AI compute

3. Concentration & fragmentation

$\text{HHI_flat} = \text{HHI_F} \times (\mu_a \mu_b + \text{alignment covariance})$, exact identity. Alignment covariance is mildly positive from 2017, peaks ~2019, decays through the AI era: no compounding-concentration signature.

AI compute (3 codes) — joint concentration decomposition (3m MA; dotted = 2023-07, 2025-04)

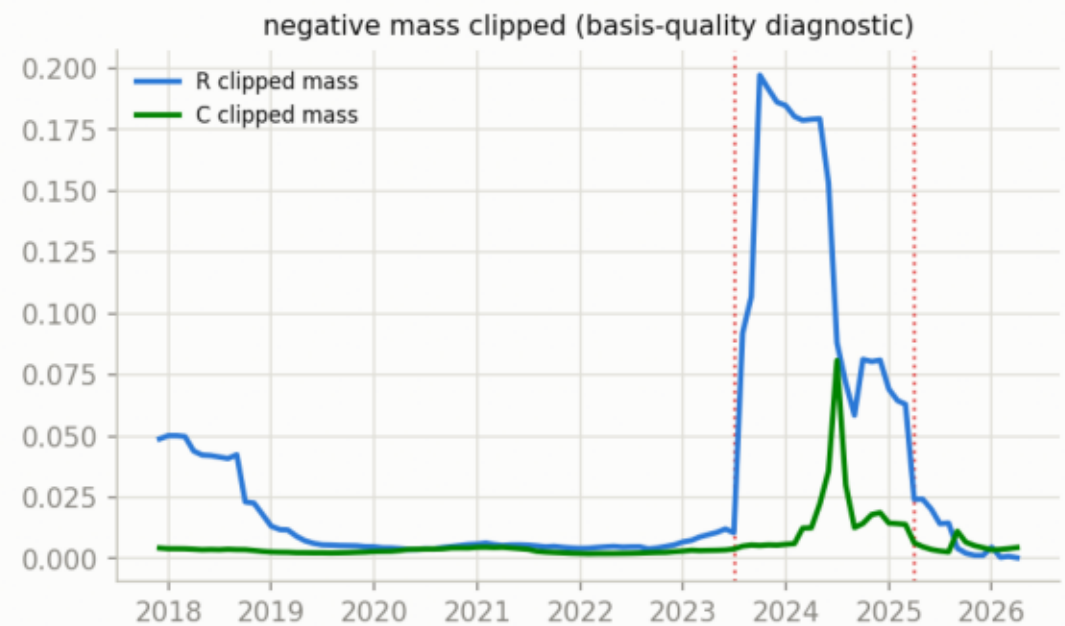
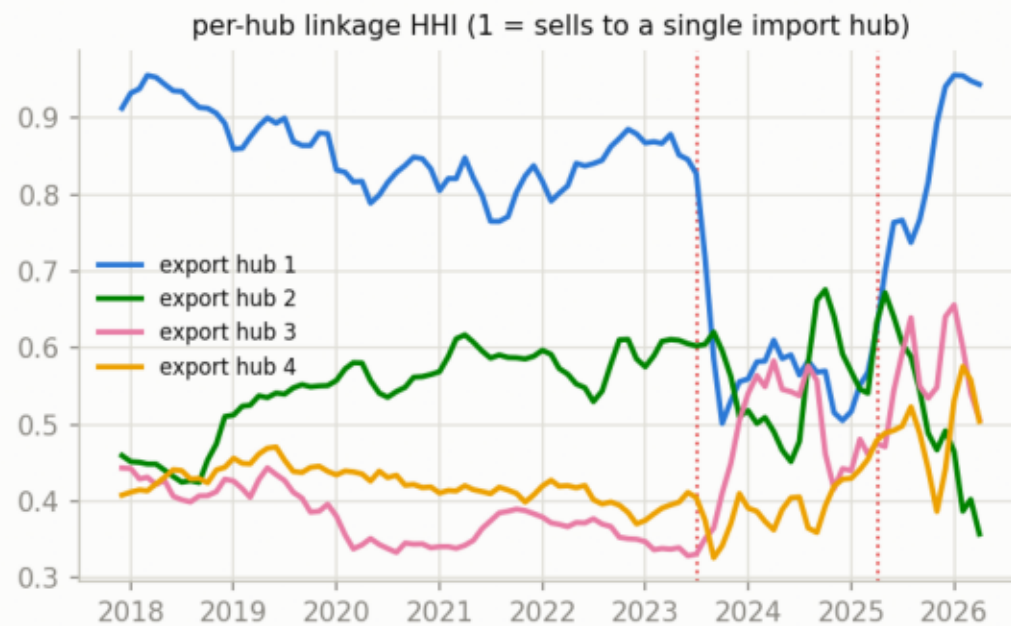


Per-hub linkage concentration and basis diagnostic -- AI compute

3. Concentration & fragmentation

Left: does an export hub sell into one import program or several. Right: clipped negative mass (basis quality), elevated during the 2023-24 regime transition.

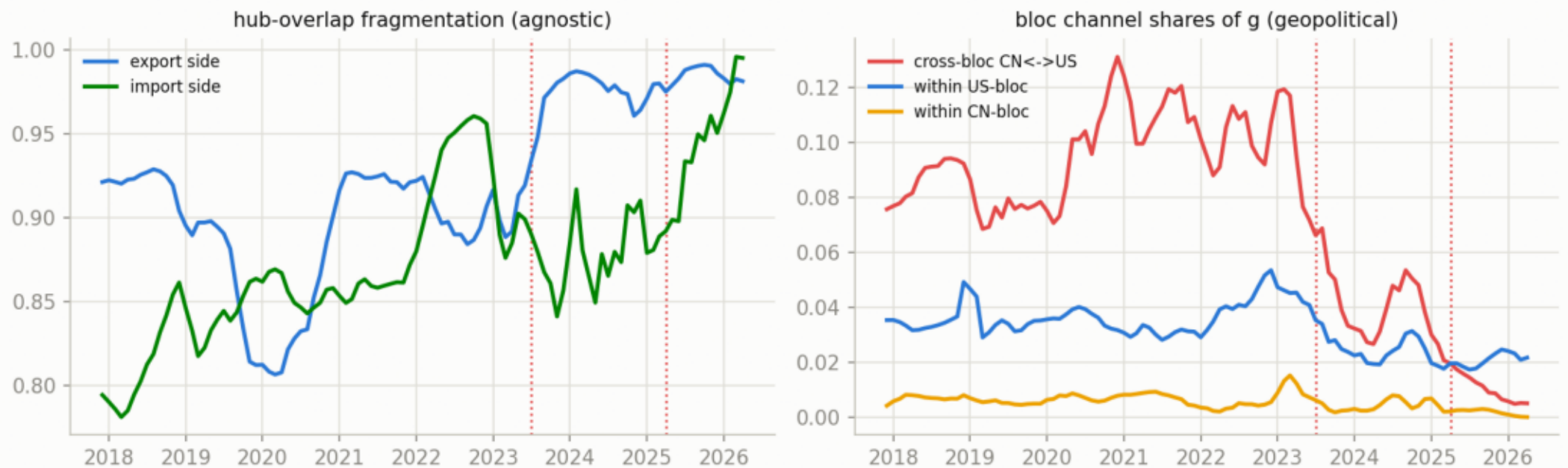
AI compute (3 codes) — linkage structure and basis diagnostics



Fragmentation -- baseboards (847180)

The sharpest decoupling of the three codes: the cross-bloc share collapses ~20x from its 2021 peak while linkage concentration HHI_F rises 0.09 -> 0.48.

HS 847180 (baseboards) — two notions of fragmentation (3m MA)



Fragmentation -- parts (847330)

Cross-bloc share halves but remains the highest of the three codes: the parts layer stays the shared substrate.

HS 847330 (parts/GPU cards) — two notions of fragmentation (3m MA)

